

Decent Coaching Classes

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Academic Conquest - II

ANSWER KEY

Subject: Science and Technology - Part 1 — Chapter 6: Refraction of Light

Q1 (A) — Choose the correct option

1. b) refraction
2. b) denser medium to a rarer medium
3. c) red
4. c) refractive index

Q1 (B) — Answer in one sentence (Any 4 of 5)

1. Refraction of light is the change in direction of light when it goes from one transparent medium to another transparent medium.
2. Dispersion of light is the process of separation of light into its component colours while passing through a medium.
3. Snell's law states that for a given pair of media, the ratio of $\sin i$ to $\sin r$ is a constant, called the refractive index (n) of the second medium with respect to the first.
4. The critical angle is the angle of incidence in a denser medium for which the angle of refraction in the rarer medium becomes 90° .
5. This happens due to atmospheric refraction — when the Sun is slightly below the horizon, its light rays reach us along a curved path through the Earth's atmosphere, so we see the Sun even before it actually rises.

Q2 — Give scientific reasons (Any 2 of 3)

1. As the density of air increases with decreasing height, its refractive index also increases, so starlight bends continuously towards the normal as it travels through the atmosphere. Due to the motion of air and changing air density and temperature, the refractive index of air keeps changing continuously, making the star's apparent position and brightness change continuously — this is seen as twinkling. Planets are much closer and appear as a collection of point sources, so the average position and brightness remain unchanged, and they do not twinkle.
2. Light coming from the part of the pencil inside water gets refracted (bent) as it passes from water (denser medium) to air (rarer medium) at the water surface, before reaching our eyes. This bending of light rays makes the submerged part of the pencil appear displaced, so the pencil appears bent at the surface.
3. Diamond has a very high refractive index (2.42), which gives it a very small critical angle. As a result, light entering a diamond undergoes total internal reflection multiple times before emerging, making it sparkle more brilliantly than ordinary glass, which has a much lower refractive index.

Q3 — Distinguish between (Any 2 of 3)

1. Refraction of light is the change in direction of a light ray when it passes from one transparent medium to another. Dispersion of light is the splitting of white light into its seven component colours (VIBGYOR) because different colours have different refractive indices in the same medium. Refraction can occur for light of a single colour; dispersion occurs only for light containing multiple wavelengths/colours, such as white light.
2. In partial reflection, part of the light incident at the boundary of a denser and rarer medium is reflected back into the denser medium while the rest is refracted into the rarer medium; this happens for all angles of incidence. In total internal reflection, when the angle of incidence exceeds the critical angle, all of the light is reflected back into the denser medium and none is refracted/transmitted into the rarer medium.
3. When light passes from a rarer medium to a denser medium, it bends towards the normal (angle of refraction is smaller than the angle of incidence). When light passes from a denser medium to a rarer medium, it bends away from the normal (angle of refraction is larger than the angle of incidence).

Q4 — Answer in detail (Any 2 of 3)

1. (i) The incident ray and the refracted ray at the point of incidence are on opposite sides of the normal to the surface at that point, and the incident ray, refracted ray, and the normal all lie in the same plane. (ii) For a given pair of media,

the ratio of $\sin i$ (angle of incidence) to $\sin r$ (angle of refraction) is a constant, called the refractive index (n) of the second medium with respect to the first — this is also known as Snell's law.

2. A mirage is caused by atmospheric refraction near a hot road or desert surface. The air near the hot surface is hotter and hence rarer than the air above it, so the refractive index of air increases with height in this region. Light rays coming from a distant object (such as the sky) bend as they pass through layers of air of increasing refractive index, and appear to be coming from an image of the object inside the ground — creating the illusion of water/reflection, called a mirage. (Diagram: shows light rays from the sky curving upward from near the hot surface towards the observer's eye, with the apparent image appearing below the actual object, near the ground.)

3. A rainbow is formed by the combined effect of dispersion, refraction, and total internal reflection of light. Small water droplets in the atmosphere act like tiny prisms. When sunlight enters a water droplet, it first gets refracted and dispersed into its component colours. The dispersed light then undergoes total internal reflection inside the droplet. Finally, the light gets refracted again as it exits the droplet. The combined effect of these three processes across many droplets produces the coloured arc of a rainbow that we see after rainfall.

Q5 — Numerical Examples (Any 2 of 3)

1. Given: $n = 1.33$, V_1 (vacuum) = 3×10^8 m/s. Using $n = V_1/V_2$: $V_2 = V_1/n = (3 \times 10^8)/1.33 = 2.26 \times 10^8$ m/s. The velocity of light in water is 2.26×10^8 m/s.

2. Given: V_2 (medium) = 2×10^8 m/s, V_1 (vacuum) = 3×10^8 m/s. Using $n = V_1/V_2$: $n = (3 \times 10^8)/(2 \times 10^8) = 1.5$. The absolute refractive index of the medium is 1.5.

3. Given: refractive index of glass with respect to air (a_{ng}) = 1.5. Refractive index of air with respect to glass (g_{na}) = $1/(a_{ng}) = 1/1.5 = 2/3 \approx 0.67$.